## SERVICE HEADER (Fill Once)

| Service ID | Service Name | Dept ID | Version | Status | Total Steps | Est. Duration |
|---|---|---|---|---|---|---|
| SVC-001 | [Building Permit] | 1 | 1 | DRAFT | 4 | 144 hrs |

## WORKFLOW STEPS MASTER TABLE (Core Configuration)

| Step | Role Code (ID) | Form Code (ID) | Jurisdiction (ID) | Strategy (ID) | Actions (IDs) | Transitions (JSON) | Assignment Rule | SLA Hrs | Escalation | Parallel OK | Notes |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | APPLICANT (3) | AF (1) | - | - | 4 | {"SUBMIT":{"next step":2}} | {} | **0** | - | ❌ | Applicant uploads |
| **2** | NODAL (7) | PF (2) | DISTRICT (2) | ROLE (1) | 1,2,3 | {"FORWARD":{"next step":3,"next_roles":[33]},"APPROVE":{"next step":4},"REJECT":{"next step":1}} | {"strategy":"ROLE","role id":7,"targets":[{"role id":33}]} | **72** | Role 15 @48h | ✅ F only | Doc verification |
| **3** | VERIFIER (33) | VF (3) | DISTRICT (2) | OFFICE (3) | 1,2 | {"FORWARD":{"next step":4},"APPROVE":{"next step":4}} | {"strategy":"OFFICE","office_ids":[101,102]} | **48** | - | ❌ | Field inspection |
| **4** | APPROVER (8) | PF (2) | STATE (1) | RULE (4) | 2 | {"APPROVE":{"next step":0}} | {"strategy":"RULE","conditions":[{"field":"amount","op":">","value":100000}]} | **24** | Dept Head | ❌ | Final decision |

## MASTER REFERENCE TABLES (Copy Codes Here ↓)

### ROLES

| Code | ID | Name |
|---|---|---|
| APPLICANT | 3 | Applicant |

| Code | ID | Name |
|---|---|---|
| NODAL | 7 | Nodal Officer |
| VERIFIER | 33 | Field Verifier |
| APPROVER | 8 | State Approver |
| DC | 15 | District Collector |

## FORM TYPES

| Code | ID | Name |
|---|---|---|
| AF | 1 | Applicant Form |
| PF | 2 | Processing Form |
| VF | 3 | Verifier Form |

## JURISDICTION

| Code | ID | Name |
|---|---|---|
| STATE | 1 | State Level |
| DISTRICT | 2 | District Level |
| BLOCK | 3 | Block Level |

## ASSIGNMENT STRATEGY

| Code | ID | Name |
|---|---|---|
| ROLE | 1 | Role Based |
| USER | 2 | Manual Users |
| OFFICE | 3 | Office Based |
| RULE | 4 | Conditional Rules |

## ACTIONS (Use IDs)

| Code | ID | Name | Color |
|---|---|---|---|
| FORWARD | 1 | 🔄 Forward | blue |
| APPROVE | 2 | ✅ Approve | green |
| REJECT | 3 | ❌ Reject | red |
| SUBMIT | 4 | 📤 Submit | orange |
| REVERT | 5 | 🔙 Revert | yellow |

## SUBFORM ACTIONS

| Code | Use |
|---|---|
| `DOCUMENT_VERIFICATION` | Docs check |
| `INSPECTION` | Field visit |

| Code | Use |
|---|---|
| `TRANSACTION` | Payment/audit |
| `""` | None |

# ADVANCED FEATURES SUPPORTED

## 1. Complex Transitions

```
{"FORWARD":{"next_step":3,"next_roles":[33,15]},"APPROVE":{"next_step":4}}
```

## 2. Assignment Rules

```
{"strategy":"RULE","conditions":[{"field":"district","op":"=","value":"Dehradun"}]}
{"strategy":"USER","user_ids":[1001,1005]}
```

## 3. SLA Escalation

| Format | Example |
|---|---|
| `@48h:Role15` | Escalate to Role 15 after 48h |
| `Head@24h` | To Dept Head after 24h |

## 4. Parallel Processing

| Symbol | Meaning |
|---|---|
| ✅ F only | Parallel FORWARD allowed |
| ✅ All | All actions parallel |
| ❌ | Sequential only |

# AUTOMATIC VALIDATION RULES

- ✅ Steps must be sequential (1,2,3...)
- ✅ Every step needs ≥1 action
- ✅ SLA auto-sum for total duration
- ✅ Role/Form codes validated from tables
- ✅ JSON syntax checked

# GENERATION COMMANDS

**Excel/Sheets → JSON:**

```
=FILL_TABLE() → workflow-SVC001-v1.json
```

**Expected Output Preview:**

```
[
  {
    "step": 1,
    "serviceId": "SVC-001",
    "roleId": 3,
    "formTypeId": 1,
    "actionMasterIds": [4],
    "slaHours": 0,
    "transitionMapJson": {"SUBMIT": {"next_step": 2}}
  },
  {
    "step": 2,
    "serviceId": "SVC-001",
    "roleId": 7,
    "formTypeId": 2,
    "jurisdictionId": 2,
    "strategyId": 1,
    "actionMasterIds": [1, 2, 3],
    "slaHours": 72,
    "transitionMapJson": {
      "FORWARD": {"next_step": 3, "next_roles": [33]},
      "APPROVE": {"next_step": 4},
      "REJECT": {"next_step": 1}
    }
  },
  {
    "step": 3,
    "serviceId": "SVC-001",
    "roleId": 33,
    "formTypeId": 3,
    "jurisdictionId": 2,
    "strategyId": 3,
    "actionMasterIds": [1, 2],
    "slaHours": 48,
    "transitionMapJson": {
      "FORWARD": {"next_step": 4},
      "APPROVE": {"next_step": 4}
    }
  },
  {
    "step": 4,
    "serviceId": "SVC-001",
    "roleId": 8,
    "formTypeId": 2,
    "jurisdictionId": 1,
    "strategyId": 4,
    "actionMasterIds": [2],
    "slaHours": 24,
    "transitionMapJson": {"APPROVE": {"next_step": 0}}
  }
]
```